

Here's the full phase roadmap we set up, with what each one delivers:

#	Phase Name	Purpose
1	Scaffolding & API Contracts	Repo structure, Pydantic schemas, and stubbed FastAPI endpoints for every module — establishes the fixed data contract every later phase builds against.
2	Browser Extension — Content Extraction	Manifest V3 extension that scrapes URL, text, headings, links, forms, images, and metadata from the active tab and sends it to the backend.
3	Content Preprocessing Pipeline	Cleans and merges raw scraped fields into one normalized text blob (HTML stripping, unicode normalization, whitespace cleanup, dedup, noise filtering) — the shared input for all NLP/ML modules.
4	AI Authorship Detection Classifier	Independent BERT-based model that predicts AI-generated vs. human-written probability — no intent judgment.
5	Malicious Intent Detection Classifier	Second, fully independent BERT-based model that classifies phishing/scam/misinformation/propaganda/impersonation/etc.
6	Claim Extraction Module	Non-classifier NLP module that identifies checkable factual claims in the text, preparing them for verification.
7	Fact Verification Pipeline	Claim Extraction → Evidence Retrieval → Evidence Comparison (entailment model) → Credibility Score.

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| 8 | Website Risk Analysis Module | Technical risk signals independent of language models: Lexical URL Analysis, HTTPS Analysis, Redirect Analysis, Suspicious Link Analysis, Source & Content Similarity Analysis. |
| 9 | Evidence Aggregation + XAI (SHAP) | Orchestrator wires together real module outputs into one evidence vector; SHAP explains each module's prediction individually. |
| 10 | Threat Scoring Engine | Transparent, weighted aggregation of the evidence vector into a final Safe → Critical Risk category. |
| 11 | Database Layer & Full Orchestration | Persistence layer (PostgreSQL/SQLite) plus the real end-to-end /analyze pipeline wiring, replacing the mocked orchestration from Phase 1. |
| 12 | Extension Report UI | Replaces the debug JSON popup with the real, user-facing threat report inside the extension. |
| 13 | Analytics Dashboard | Frontend + backend query endpoints for all the analytics categories (threat trends, AI-content stats, category breakdowns, website risk, explainability stats). |
| 14 | Integration Testing & Refinement | Full end-to-end pipeline pass, edge-case handling, and polish across every module. |